

Structured bilevel optimization with many follower variables: Global responses, accuracy, and structural boundaries

1 Problem setting and foundations

Many leader–follower models contain a large number of local follower choices but only a few shared resources, aggregate performance measures, and leader decisions. A tariff may influence the operation of many units, while their nonlinear interaction depends only on total load or emissions. The relevant question is whether the number of shared quantities can control the complexity of global bilevel optimization even as the number of follower variables grows. The distinction between the dimension of the follower and the dimension needed to describe its responses is central.

A tractable follower problem alone does not answer this question. The leader optimizes over the responses induced by all its possible decisions; this optimization can be nonconvex even if every follower has a unique convex minimizer. Conversely, nonconvexity confined to a fixed number of aggregate coordinates need not prevent an exact description of all global responses. The development below separates three tasks: describing global follower responses, optimizing over those responses under a specified selection convention, and representing the requested output at a stated accuracy.

1.1 Related work and the structural question

Classical positive results for linear bilevel optimization fix the number of *follower* variables. Liu and Spencer [10] give a polynomial algorithm under that restriction; Deng [5] discusses the resulting complexity theory. This restriction differs from fixing the number of leader variables while allowing the follower dimension to grow. Indeed, Sugishita and Carvalho [12] prove NP-completeness with one leader variable, unit-bounded variables, and no additional upper constraints for a linear bilevel class with coupled follower constraints. Their polynomial algorithm for finding a local optimum does not give global tractability.

Ketkov and Prokopyev [8] distinguish fixed follower variable and constraint counts, optimistic and pessimistic responses, and linear and quadratic models. Their optimistic convex-quadratic positive result fixes the follower dimension and imposes their stated convex quadratic upper structure. Neither that result nor a restriction on the total follower constraint count encompasses a model with arbitrarily many local follower variables and local bounds, few shared constraints, and a possibly nonconvex polynomial aggregate cost. In particular, counting shared resource rows separately from local rows is essential. Their pessimistic convention also illustrates why upper feasibility must be specified before complexity statements are compared.

The local elimination tools have direct antecedents. Megiddo and Tamir [11] use low-dimensional multiplier arrangements for separable convex quadratic optimization and for fixed-dimensional quadratic blocks with shared constraints. The local quadratic formulas and multiplier partitions used here build on this principle. Exact sign determination and quantifier elimination in fixed dimension are classical real algebraic algorithms [1]. The structural question is how these ingredients

provide a description of *globally* optimal responses when the aggregate cost is nonconvex, and how that description supports the upper optimization. Stationarity alone cannot supply this last step.

For polynomial local costs, earlier work already supplies high-accuracy methods for separable resource allocation [6] and approximation schemes for sums of algebraic functions [13]. A high-accuracy follower evaluation at a fixed leader is different from a global approximation algorithm for the induced, possibly signed upper objective. Both the numerical degree and the accuracy dependence must be tracked in making that comparison.

Cost-near-optimal follower behavior is an established form of decision uncertainty. Besançon, Anjos, and Brotcorne [3] study robustness of upper feasibility against such responses; Beck, Ljubić, and Schmidt [2] place this model in the broader uncertainty literature. The robust model below also allows a worst-case upper objective. The structural contribution concerns exact compression of its adversarial problems, including nonconvex nominal followers, rather than the introduction of near-optimality robustness.

Finally, polynomial-size optimality certificates need not produce polynomial-time optimization. Buchheim [4] proves NP membership for linear bilevel optimization and polynomial encoding bounds for valid big- M values in optimistic KKT reformulations. Practical reformulations, branching, cuts, and decomposition are surveyed by Kleinert et al. [9]. They provide the computational context for the specializations studied here. Polynomial time for fixed structural dimensions is a worst-case statement; it is not by itself a claim of favorable numerical performance.

1.2 The quadratic-block model

The leader chooses $x \in C \subset \mathbb{R}^r$, where C is a compact set specified by a rational quantifier-free polynomial formula. The follower vector is $z = (z_1, \dots, z_B)$, with $z_b \in \mathbb{R}^{d_b}$, $d_b \leq d$, and total dimension $N = \sum_{b=1}^B d_b$. Both B and N may grow. For the primary model, let

$$P_b(x) = \{y \in \mathbb{R}^{d_b} : E_b y = e_b(x), G_b y \leq h_b(x)\}, \quad (1)$$

$$P(x) = \{z : z_b \in P_b(x) \ (b = 1, \dots, B), \ Az = b(x), \ Hz \leq a(x)\}. \quad (2)$$

The matrices E_b, G_b, A, H are rational and independent of x . There are k_+ shared equality rows and k_- shared inequality rows, and $k = k_+ + k_-$. The number of local rows is unrestricted. All right-hand sides are polynomial in x . Among the local inequalities are coordinate bounds $\ell_{bi}(x) \leq z_{bi} \leq u_{bi}(x)$, with polynomial endpoints satisfying $\ell_{bi}(x) \leq u_{bi}(x)$ on C . Thus the union of the follower feasible sets is bounded. A set $P_b(x)$ or $P(x)$ may be empty; follower feasibility is not a promise at every leader.

Let $U(x) \in \mathbb{Q}[x]^{s \times N}$, and partition its columns into blocks $U_b(x)$. The follower objective is

$$f(x, z) = \sum_{b=1}^B \left(\frac{1}{2} z_b^\top Q_b(x) z_b + c_b(x)^\top z_b \right) + \phi(x, U(x)z). \quad (3)$$

Each $Q_b(x)$ is a symmetric polynomial matrix positive definite for every $x \in C$. The polynomial $\phi(x, w)$ can be nonconvex in w . Its argument $w = U(x)z \in \mathbb{R}^s$ collects the shared aggregate quantities. Positive definiteness is imposed on the local blocks, not on the entire follower Hessian. In particular, the follower need not have a unique minimizer.

The upper objective $F(x, z)$ and upper rows $g_j(x, z) \leq 0$ ($j = 1, \dots, m$) are explicitly encoded rational polynomials. Their monomials may involve many different follower coordinates; neither m nor their support sizes are structural parameters. Unless a theorem says otherwise, the fixed parameters of the exact quadratic model are

$$(r, d, s, k_+, k_-). \quad (4)$$

The scalar local model has $d = 1$ and only interval constraints within a block. Extensions with polynomially varying constraint normals will be stated separately. The approximation results use strictly convex univariate polynomial local costs and a convex aggregate cost, with their own leader-domain and data assumptions; those assumptions are not implicit extensions of (3).

1.3 Follower selection and upper feasibility

For $P(x) \neq \emptyset$, compactness gives the value and solution set

$$v(x) = \min_{z \in P(x)} f(x, z), \quad S(x) = \{z \in P(x) : f(x, z) = v(x)\}. \quad (5)$$

Put $S(x) = \emptyset$ if $P(x) = \emptyset$, and use $v(x)$ only on feasible leaders. The follower always optimizes over $P(x)$ without upper constraints. Define the optimistic feasible set and value by

$$\mathcal{O} = \{(x, z) : x \in C, z \in S(x), g_j(x, z) \leq 0 \ (j = 1, \dots, m)\}, \quad (6)$$

$$\text{val}_{\text{opt}} = \inf_{(x, z) \in \mathcal{O}} F(x, z). \quad (7)$$

Thus the leader may select a favorable global response that satisfies its upper rows. A minimizer is asserted only when attainment is established.

For pessimistic optimization, upper feasibility is universal:

$$D_0 = \{x \in C : S(x) \neq \emptyset, g_j(x, z) \leq 0 \text{ for all } z \in S(x), j = 1, \dots, m\}, \quad (8)$$

$$W_0(x) = \max_{z \in S(x)} F(x, z), \quad \text{val}_{\text{pes}} = \inf_{x \in D_0} W_0(x). \quad (9)$$

The maximum at a fixed feasible leader is attained. The infimum over leaders need not be. An upper constraint cannot discard an inconvenient follower optimum before W_0 is computed.

Given a nonnegative polynomial budget $\rho(x)$ on C , the set of cost-near-optimal responses is

$$S_\rho(x) = \{z \in P(x) : f(x, z) \leq v(x) + \rho(x)\} \quad (10)$$

at feasible leaders, and is empty otherwise. Replacing $S(x)$ by $S_\rho(x)$ in (8)–(9) defines D_ρ , W_ρ , and val_ρ . A bounded budget may itself be a leader coordinate when the leader model permits it. Near-optimal points need not satisfy follower stationarity. The robust compression theorem will restrict the number of affine measurements of z used by each upper criterion; this is an additional hypothesis, distinct from the unrestricted polynomial upper data in the exact optimistic model.

For revenue maximization, the corresponding optimistic objective is a supremum and the pessimistic objective maximizes the minimum revenue over all responses. Equivalently, negate revenue in the minimization conventions above. All algorithms report infeasibility separately, and state whether the finite optimum value is attained.

1.4 Input length, degree, and output

The bit model counts arithmetic on binary-encoded rational data. Let L be the total rational input encoding length, including dimensions, row indices, monomial lists, exponents, and the Boolean formula for C . Let $\delta \geq 2$ bound the actual total degree of every input polynomial. The exact algorithms have running time polynomial in (L, δ) for fixed (4). A claim of polynomial time in L alone assumes that δ is polynomially bounded by L , as for dense or unary-exponent encodings. A sparse monomial representation with unrestricted binary exponents does not have that property.

Polynomial local costs in the approximation results are supplied densely; a power-specific result will state its dependence on the numerical power.

A common-field algebraic output consists of a square-free polynomial $p \in \mathbb{Z}[T]$, a rational isolating interval selecting a real root α , and rational polynomials p_i, q_i such that each returned coordinate equals $p_i(\alpha)/q_i(\alpha)$ with $q_i(\alpha) \neq 0$. The guarantee includes the degree of p , all coefficient bit lengths, and the combined encoding of all coordinates. They are bounded polynomially in the input size in the stated fixed dimensions. The polynomial p need not be a minimal polynomial. It suffices that its chosen root specifies the same real extension for every recovered coordinate. If several adversarial problems are requested independently, a common-field guarantee for each leader–witness pair does not imply a polynomial-degree field containing all their unrelated witnesses at once.

An accuracy-bit guarantee takes a positive rational tolerance ε , whose encoding contributes to the running time. Writing $\varepsilon = 2^{-\beta}$ illustrates the distinction between a bound polynomial in β and one polynomial in $1/\varepsilon$. In the base approximation model, the output is a rational leader that is exactly feasible for the base leader domain and admits a feasible follower, with an additive guarantee for the true induced upper objective. An approximate follower vector does not replace the true response in that guarantee. Exact satisfaction of response-dependent upper rows requires additional conditions; without them, a theorem must specify its allowed upper violation. Exact algebraic output, rational near-optimal output, and numerical evaluation are different computational tasks.

The exponents of the polynomial bounds depend on the structural parameters. For the decision versions this gives XP-type bounds, $L^{h(r,d,s,k)}$, under the degree convention. It does not establish a fixed-parameter bound $h(r,d,s,k)L^c$ with an absolute exponent c .

1.5 Algebraic and polyhedral tools

We use the following standard consequences of fixed-dimensional real algebraic computation [1, Theorem 1.3.1 and Section 3.1.3]. For a rational polynomial family in a fixed total number of real variables, realizable sign conditions, including zero signs, can be enumerated in time polynomial in the number of polynomials, their degrees, and coefficient bit lengths. A first-order formula with a fixed total number of free and bound variables can be replaced by an equivalent quantifier-free formula within the same kind of bound. Nonempty semialgebraic sets admit algebraic samples of polynomial degree and encoding length. Applied simultaneously to all free coordinates, sampling gives the common-field representation described above. The total number of variable occurrences introduced by quantified copies must be controlled; fixing only the number of free variables is insufficient.

One elementary encoding fact will be used repeatedly.

Lemma 1.1 (Substitution in fixed dimension). *Let $v \in \mathbb{R}^q$, with q fixed, and suppose that $Z_1(v), \dots, Z_N(v)$ and $D(v)$ have explicit polynomial encodings of polynomial size and degree. On a region where $D > 0$, put $z_i = Z_i/D$. Let $p(x, z)$ be an explicitly listed rational polynomial of total degree at most δ , with the coordinates of x among those of v . Then*

$$D(v)^\delta p(x, Z(v)/D(v))$$

is a polynomial that can be constructed in time polynomial in the input encoding lengths, the degrees of Z_i, D , and δ . Its sign agrees with that of $p(x, z)$ on the region.

Proof. An input monomial $ax^\nu z^\gamma$ becomes $ax^\nu \prod_i Z_i^{\gamma_i} D^{\delta-|\gamma|}$. Its degree is at most $\delta(1+T)$ if the degrees of Z_i, D are at most T . A polynomial of degree at most $\delta(1+T)$ in q variables has at most

$\binom{\delta(1+T)+q}{q}$ monomials. This number is polynomial for fixed q . Expanding and adding the explicit input monomials therefore takes polynomially many rational operations; coefficient bit lengths grow at most polynomially under these products and sums. Positivity of D^δ proves sign preservation. $\square$

For polyhedral limits we use Hoffman’s bound [7]. Given fixed matrices M, T , there is a constant $H_{M,T}$ such that, whenever $K(a, b) = \{y : My \leq a, Ty = b\}$ is nonempty,

$$\text{dist}(y, K(a, b)) \leq H_{M,T}(\|(My - a)_+\| + \|Ty - b\|). \quad (11)$$

Any fixed choices of finite-dimensional norms can be used, with a changed constant. The constant is uniform in the right-hand sides. Equalities can be encoded as opposite inequalities. In the exact attainment proof, only existence of this constant for each instance is needed. The approximation proofs require quantitative encoding bounds and will establish them explicitly. Neither form of the argument grants a uniform constant when the constraint normals vary with the leader.

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